

Mohak Subramanya Murthy

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Technologies

Languages: Java, Python, C++, JavaScript, TypeScript, HTML, CSS

Frameworks: React, Node.js, Express.js, Next.js, Flask, FastAPI

Databases: SQL, MongoDB

Machine Learning Libraries: TensorFlow, PyTorch, Keras, JAX, Scikit-learn, Numpy, Pandas, Matplotlib, OpenCV

Dev Tools: Git, Github, MLFlow, Unix, Docker, Vim

Education

VIT Bhopal University - Computer Science Engineering in AI and ML

October 2022 - Present

- CGPA: 8.88/10

St. Francis de Sales Public School (ISC)

June 2020 - May 2022

- Percentage: 91%

St. Francis de Sales Public School (ICSE)

June 2008 - May 2020

- Percentage: 94.6%

Publications

- **Subramanya Murthy, M., Shetty, D.P., "Rethinking Temporal Graph Networks: A Comparative Study for Social Network Analysis."** Accepted at IEEE PUNECON 2025, Accepted at IEEE ICCT 2025 (withdrawn prior to presentation)

Work Experience

NITK Surathkal - Research Intern

May 2025 - June 2025

- Developed Temporal Graph Networks (TGN) with fixed and adaptive decay mechanisms, enhancing community detection and link prediction on a Reddit dataset with 850K+ timestamped interactions, achieving a 1.6% improvement in silhouette score and 0.2% increase in MRR@100.
- Identified trade-offs in decay strategies, with adaptive decay showing enhanced performance in sparse node settings and the ability to balance recency and long-term memory retention, leading to task-specific improvements in predictive accuracy.

Omdena AI - Junior ML Engineer

March 2025 - June 2025

- Utilized tools such as Python, PyTorch, OpenCV, GDAL, rasterio, and ACOLITE to build an end-to-end geospatial ML pipeline for detecting plastic debris across the Italian and Mediterranean Seas, enabling scalable, automated monitoring and reducing manual analysis time by 70%.
- Preprocessed Sentinel-2 satellite imagery using ACOLITE for atmospheric correction and performed spectral analysis to distinguish plastic debris from natural water features, generating robust inputs for model training.
- Developed and fine-tuned an Attention U-Net model for semantic segmentation, achieving 85% precision and 78% recall, with performance enhanced through custom spectral indices, loss function optimization, and hyperparameter tuning on the MARIDA dataset.

Projects

Diabetic Retinopathy Analysis with Grading and Lesion Segmentation

[Link](#)

- Tools Used: Python, PyTorch, Pandas, Scikit-learn, Optuna, OpenCV
- Developed an end-to-end, multi-task medical imaging pipeline leveraging EfficientNetV2-S for simultaneous diabetic retinopathy severity classification (5-class) and lesion segmentation, using stratified data sampling and progressive training strategies. Achieved robust clinical performance (accuracy 85.5%, Dice score 74.1%) and optimized deployment for resource-constrained GPUs.
- Applied advanced data augmentation, feature extraction, and cross-validation techniques on APTOS 2019 and IDRiD datasets, with rigorous model evaluation (sensitivity, specificity, kappa, Dice coefficient) and clear documentation for reproducibility and research compliance.

Indian Sign Language Recognition

[Link](#)

- Tools Used: Python, Pandas, Scikit-learn, Tensorflow, Keras, Mediapipe, OpenCV, Flask
- Developed an AI-powered system for real-time translation of Indian Sign Language (ISL) to text and speech, leveraging machine learning and deep learning to enhance accessibility and enable seamless communication for the deaf and hard-of-hearing community.
- Created and optimized a custom ISL gesture dataset through video recordings, data augmentation, and transfer learning, achieving high-accuracy recognition across diverse signing styles and environmental conditions.